

PRABHUDAYAL VAISHNAV

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WORK EXPERIENCE

• Research Internship : ESIEA Paris, France (Remote)

January 2025 - Present

Working on the "Complex Human Emotion Detection Using YOLO11" research project.

- Enabled real-time detection of 22 discrete human emotions by training on the CFEE dataset using YOLO11.
- Fine-tuned YOLO11 (COCO-pretrained) using transfer learning for real-time emotion detection.

PROJECTS

Career Scout: Voice AI-Powered Job Search Assistant

Next.js, React, Supabase, TypeScript, MCP Server

Developed a full-stack Voice AI Career Coach for smart job search



- Architected a custom MCP server with 3 tools: *Intelligent Job Search*, *Company Analysis*, and *Web Search*.
- Integrated custom scoring logic to match user skills with confidence percentages.
- Built a responsive, Supabase-authenticated frontend with React hooks and Tailwind CSS, reducing voice interaction latency by 40%.

• The AI Scientist: Sakana AI

Python, LLMs, Transformers, Git

Contributed to Sakana AI's Open Source project to automate scientific research.



- Added 8 open-source language models, achieving a 75% reduction in generation costs.
- Adapted the project for T4 GPU support, reducing reliance on high-end GPUs like H100/A100.
- Co-authored a paper: "Exploring Style Transfer with Small Character-Level Transformers" using Qwen2.5(72B).

• Development of Python Module: XS

Python, Ngrok, API Creation

Published a pip package to expose inference endpoints for open-source LLMs.



- Released a Python package to create accessible endpoints for over 99% of Hugging Face-hosted LLMs.
- Leveraged Ngrok to tunnel requests from localhost to the cloud securely.

• Remote Shell Access Bot: XAdmin

Python, Telegram, Networking, Operating Systems

Built a Telegram-controlled bot for remote shell operations on Windows hosts.



- Enabled covert remote shell access to host PCs through Telegram commands.
- Implemented features like screenshot capture, remote file sharing, terminal access, IP tracing, and disk/file listing.

TECHNICAL SKILLS

• Programming Languages:

Python, Next.js, React, JavaScript, C, C++, SQL, Matlab, R

• AI and LLM Tools/Frameworks:

LangChain, LlamaIndex, Ollama, HuggingFace Transformers, PyTorch, TensorFlow

• Data Analysis and Visualization Tools:

Pandas, Numpy, Seaborn, Power BI

• Miscellaneous Skills:

Linux, MacOS, Competitive Programming

• Soft Skills:

Leadership, Communication, Problem Solving

RELEVANT COURSEWORK

• Core Computer Science:

Object Oriented Programming, Database Management System, Computer Architecture Organization, Computer Network, Operating System, Data Structures and Algorithms

• Artificial Intelligence and Data Science:

Artificial Intelligence, Data Visualization, Pattern Recognition and Machine Learning, Natural Language Processing

• Mathematics and Theoretical Foundations:

Theory of Computation, Cryptography and Network Security

ACHIEVEMENTS

Indian Institute of Management (Data Analyst) Hackathon Winner, 2024

2024

8 hour long Hackathon, organized by InFED at Nagpur on 12 November 2024



- Won 1st place by presenting a solution to the Startup's Problem: **AI Product Comparison**.
- Implemented the Deep-Reserach & generated a report on **AI Product Comparison (22-page report)** using a Python-based AI Agent with Tavily Search, significantly reducing manual effort and time by 90%.

EDUCATION

Bachelors of Technology (Hons.) CSE Data Science

2022-Present

(6th Semester) Chhattisgarh Swami Vivekanand Technical University, Bhilai, 491107

CGPA : 7.2/10

- Research Internship at ESIEA Paris, France (Remote) under the project "Complex Human Emotion detection with Ultralytics YOLO11".